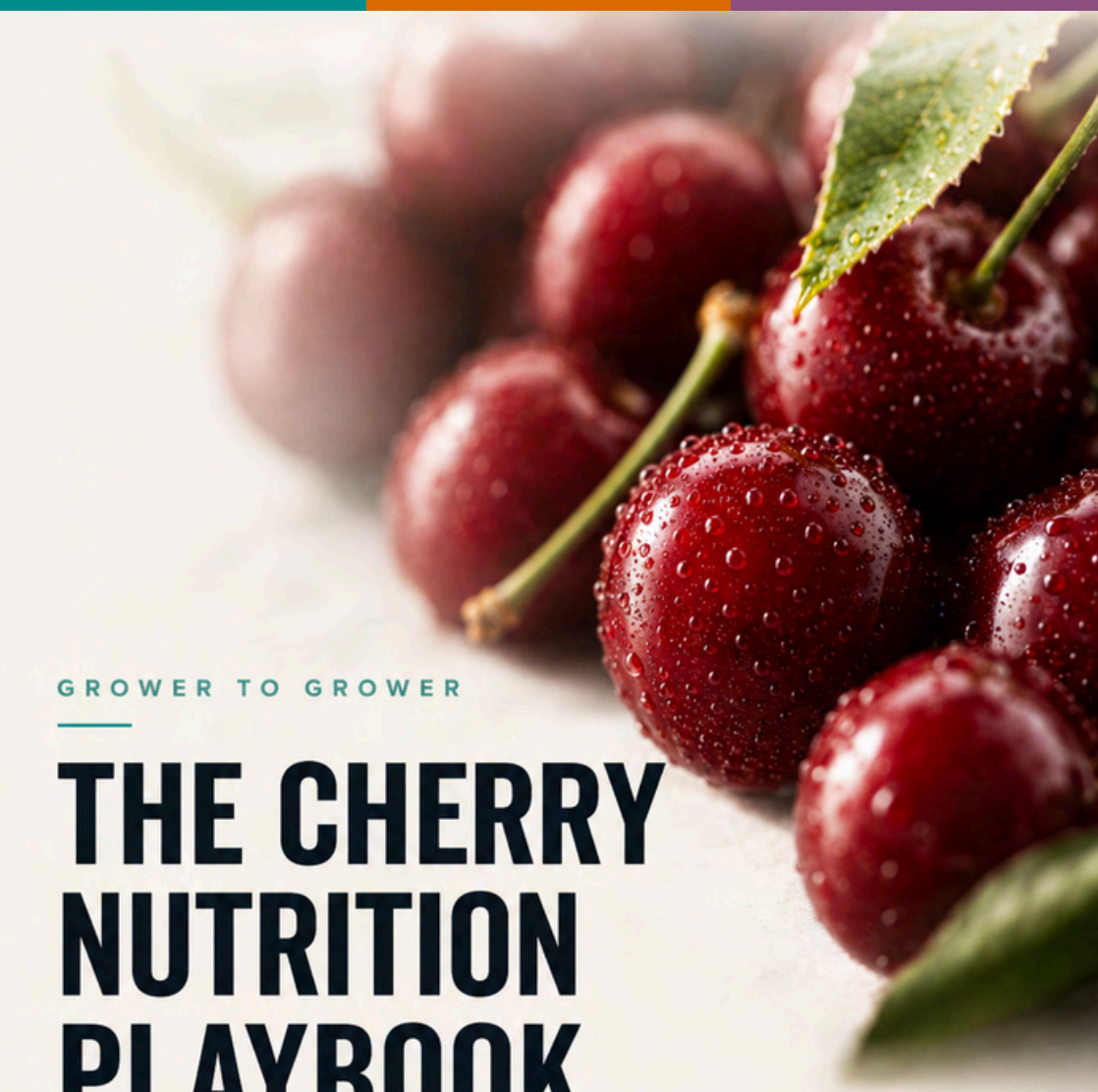




GROWER TO GROWER

THE CHERRY NUTRITION PLAYBOOK

*Physiology, testing, and seasonal levers
from a grower who's walked every one.*

HYBRID AG
PRECISION NUTRITION



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Nutritional Agronomist
& Cherry Grower



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Agronomist



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Nutritional
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HORTICULTURE TEAM LEADER
HYBRID-AG



SIXTH-GENERATION GROWER

From Smith's Fruit in Taminick,
Victoria.



140+ YEARS OF HISTORY

Smith's Fruit represents more
than 140 years of family fruit-
growing history.



GROWER & AGRONOMIST

Combining practical orchard
experience with plant physiology
to build better programs.



CHERRY NUTRITION PLAYBOOK

A practical, grower-led guide
focused on the decisions that
drive next season's crop.

A Grower Who also Builds the Programs

Most growers see harvest as the finish line. But through years of experience as both a cherry grower and agronomist, Andrew Smith has learned that the foundations of next season are built during post-harvest.

Andrew is a sixth-generation fruit grower from Smith's Fruit in Taminick, Victoria, and serves as Horticulture Team Leader at Hybrid-Ag. Smith's Fruit represents more than 140 years of family fruit-growing history.

The Cherry Nutrition Playbook brings together practical orchard experience and plant physiology to help growers better understand the seasonal levers that drive performance.

From post-harvest reserve building through to flowering, fruit set, cell division, and fruit development, this guide focuses on the decisions that shape next season's crop long before harvest arrives.

Built from both the orchard and the field, it is a practical, grower-led resource grounded in real-world cherry production and the science that supports it.

Three Blocks – One Story

Post-harvest is not the quiet end of the season. It is where much of next season begins. This playbook looks at the misconceptions, the science, and the practical levers that shape what happens next.



BLOCK 01 – COMMON MISCONCEPTIONS

Challenge the assumptions

Reframing the assumptions that influence timing, nutrition, and decision-making in cherries especially around post-harvest, chill, frost, soil activity, and testing.



BLOCK 02 – THE SCIENCE

Understand what the tree is doing

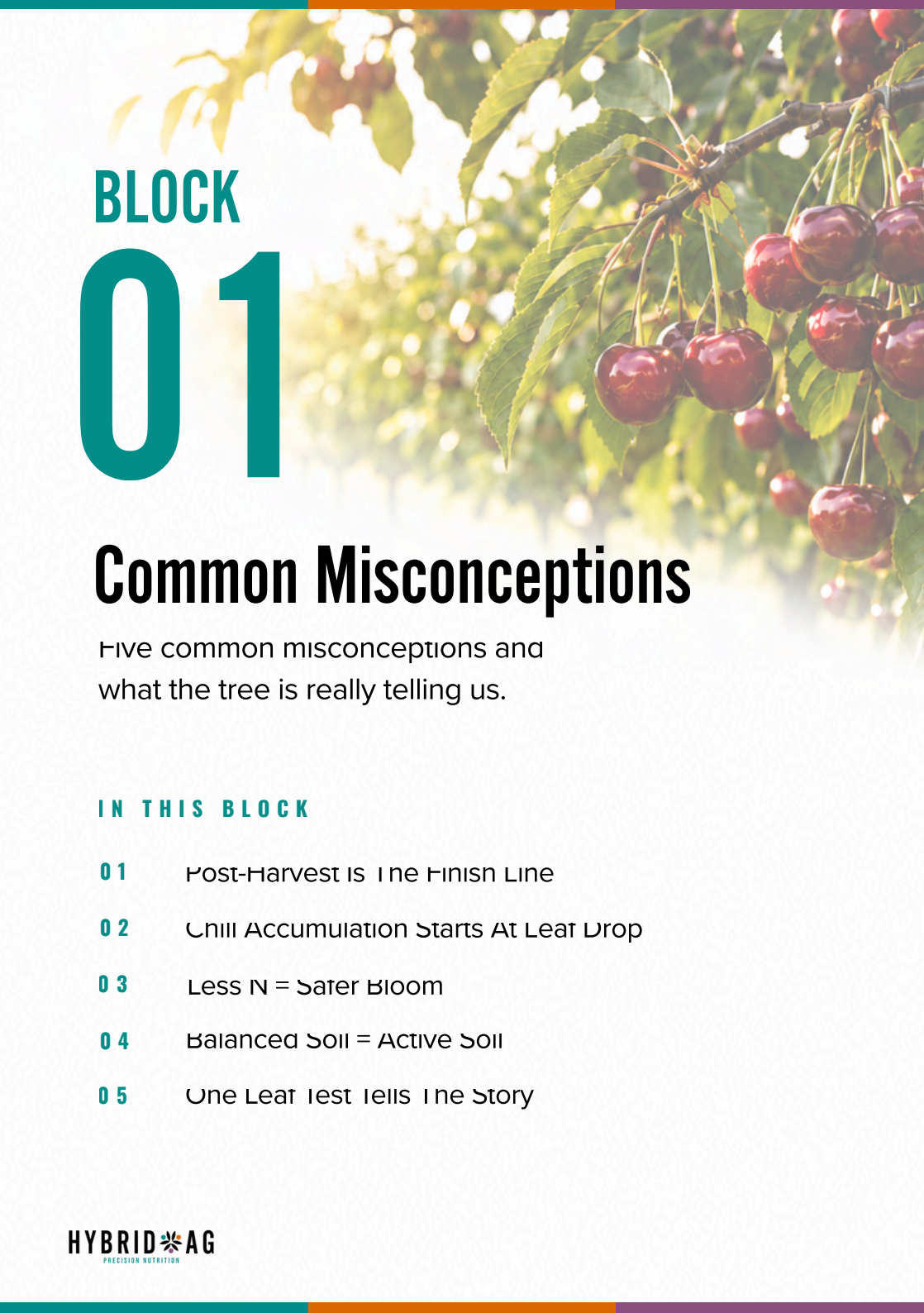
Understanding what the tree is doing after harvest: how reserves are rebuilt, nutrients are moved, buds are influenced, and spring performance begins to take shape.



BLOCK 03 – THE PLAYBOOK

Turn insight into action

Turning that understanding into practical action through the timing windows, nutrition levers, and decisions that help support stronger crop loads, better fruit quality, and more consistent outcomes.



BLOCK 01

Common Misconceptions

Five common misconceptions and what the tree is really telling us.

IN THIS BLOCK

- 01 Post-Harvest IS The Finish Line
- 02 Chill Accumulation Starts At Leaf Drop
- 03 Less N = Safer Bloom
- 04 Balanced Soil = Active Soil
- 05 One Leaf Test Tells The Story

Misconception #1 - “Post-Harvest is The Finish Line”

The fruit is off. The pressure shifts. The season can feel like it is winding down.

But for the tree, one of the most important setup phases of the year is only just beginning.

Post-harvest is when reserves are rebuilt. It is the period where bud and flower quality begin to be shaped, and where the foundation for spring growth starts to form.

The reserves a tree relies on next August are largely the reserves banked during post-harvest. If this window is missed or underestimated, the impact does not stay in post-harvest — it carries forward into the following season.

Poor reserve accumulation affects the tree’s ability to support bud break, flowering, fruit set, and early cell division. And when reserves are repeatedly run down year after year, the next season often starts behind before it has even begun.

Post-harvest is not the end of the season. It is the starting point for the next one.



Misconception #2 - “Chill Accumulation Starts at Leaf Drop”

A few seasons ago at a field day, a grower asked a simple question:

“When does chill accumulation start if the tree still has green leaves?”

The short answer is: it already has.

Endodormancy begins inside the bud before the canopy fully senesces. It is driven by changing day length, temperature, and hormonal signals within the tree. While the leaves may remain green for weeks, the bud’s internal clock is already shifting toward dormancy.

That changes the way late-season nutrition needs to be viewed.

Less-mobile nutrients such as zinc, boron, and calcium need to be loaded into buds and storage tissues before leaf drop. Once the canopy begins shutting down and transport slows, the opportunity to move those nutrients efficiently into developing buds becomes increasingly limited.

The tree may still look active from the outside, but physiologically, the transition toward next season has already begun.



**THE CHILL CLOCK
IS ALREADY TICKING**

Endodormancy begins
inside the bud — before
leaf drop.

ENDODORMANCY: THE TREE'S INTERNAL PAUSE

Growth is stopped from within, even when conditions outside are favourable.

WHAT'S HAPPENING



The bud's internal clock shifts toward dormancy.



Day length, temperature, and hormonal signals trigger endodormancy.



Growth is restricted until the bud has received enough chill.



THE CHILL CLOCK

Inside the bud, metabolic activity slows and growth signals are suppressed. THE CLOCK IS ALREADY TICKING.

THE CHILL CLOCK

1



GREEN LEAVES

The canopy may still look active, but internal signals are already shifting.

2



BUD ENTERS ENDODORMANCY

Changing day length, temperature, and hormones start the chill process inside the bud.

3



LEAF DROP

By this point, transport is slowing and the nutrient-loading window is closing.

THE DORMANCY CYCLE

CHILL ACCUMULATION

Exposure to cold satisfies the bud's chill requirement.



ACTIVE GROWTH

Bud grows and develops.

ENDODORMANCY INDUCTION

Internal signals cause the bud to shut down.

DEEP DORMANCY

Growth is locked. Bud requires chill to reawaken.



WHY TIMING MATTERS

Less-mobile nutrients need to be loaded before leaf drop, while the canopy is still moving them efficiently into buds and storage tissue.

Zn

= bud development

B

= reproductive support

Ca

= tissue strength



KEY TAKEAWAY

If you wait for leaf drop, part of the loading window has already passed.



CHILL IS EARNED, NOT INSTANT.

Warm days don't cancel dormancy, but chill doesn't accumulate when it's warm.



THE TREE LOOKS QUIET, BUT IT'S PREPARING.

Bud scales tighten, metabolism slows, and growth is paused.



THE TRANSITION HAS ALREADY BEGUN.

Dormancy starts before leaf fall and continues through winter.



YOUR ACTIONS NOW SHAPE NEXT SEASON.

Nutrition, timing, and management during this window drive stronger buds, better bloom, and higher yield.

Misconception #3 - “Less N = Safer Bloom”

Half right, half wrong. Free nitrate during a frost window can increase risk by driving softer, more succulent tissue — and softer tissue is often the first to freeze. That part is true.

What is often missed is the role nitrogen plays earlier in the season. Sweet cherry spring growth relies heavily on nitrogen reserves stored during the previous season. Much of the early growth push — including bud break, flowering, and initial shoot growth — is fuelled by reserves accumulated during post-harvest.

That means post-harvest nitrogen is not the enemy. It is one of the key tools used to rebuild reserves and support next season’s development. The real lever is not simply nitrogen volume. It is nitrogen form, timing, and conversion within the tree.

Nitrate nitrogen applied during the post-harvest phase needs to be converted into amino acids and protein forms for complete storage within woody tissues, buds, and roots. Nutrient balance, carbohydrate availability, nitrogen form, and application timing all influence how efficiently this conversion and storage process occurs.

Post-harvest nitrogen, whether soil-applied or foliar, helps replenish reserve pools. Balanced nutrition during this period supports the conversion of free nitrogen into stable storage compounds. Ammonium-based nitrogen around bud movement aligns more closely with active tree demand. Free nitrate nitrogen, however, is best kept well clear of the frost-risk window where excessive soft growth can increase vulnerability.

Timing changes the outcome. When a frost event is approaching, a balanced foliar containing calcium, potassium, and key micronutrients applied 24–48 hours before the event can help provide a measurable buffer through osmotic regulation, membrane stability, and cell-wall support. Different nutrient. Different timing. Completely different result.



BEFORE THE FROST EVENT

The frost event reveals the condition the tree carried into it.



BUD MOVEMENT • WEEKS OUT

Use nitrogen in the right form and timing.

Ammonium-based nitrogen as the tree comes out of dormancy supports early growth. Timing is critical to match tree demand and minimize soft growth.



FROST EVENT

Preparation has already shaped the tree's condition.

The tree can only perform as well as the condition it was grown into.

1

POST-HARVEST • MONTHS OUT

Rebuild carbohydrate and nutrient reserves.

Photosynthesis is still active, and this is the time to rebuild reserves for next season.

Focus on leaf health, balanced nutrition and reserve building to set up strong growth.

2



3

24–48 HOURS PRE-EVENT

Increase plant resilience to frost.

Apply a balanced foliar containing Calcium, Potassium, Phosphorus and key micronutrients 24–48 hours before the forecast frost. e.g. Frost-Ex or as advised.

4



KEY TAKEAWAY

Different nutrient program.
Different timing.
Different result.

Post-harvest nutrition builds reserves.
In-season timing supports early growth.
Pre-frost nutrition builds resilience.
The frost event reveals the work done before it.

HYBRID*AG
PRECISION NUTRITION

Misconception #4 - “Balanced Soil = Active Soil”

Balanced on paper does not always mean active in the root zone. A soil test may show strong nutrient levels and good balance, but the real question is whether the tree can actually access those nutrients when demand is high. Nutrient availability is influenced by far more than total nutrient levels alone. Biology, carbon, moisture, pH, soil structure, temperature, and root activity all affect whether nutrients move, cycle, and become plant available throughout the season.

Biology and carbon are not silver bullets, but they are important parts of the system. They help drive nutrient cycling and mineralisation that converts nutrients from “present on paper” into forms that can be accessed within the root zone.

That is why soil testing is only the starting point. A good soil test establishes the baseline and helps expose the constraints within the system. The goal is not simply to apply more fertiliser.

The goal is to understand: what the soil can supply, what it can retain, what it can release, and how efficiently it can deliver nutrients in time with tree demand. Because ultimately, the outcome is not better numbers on a report. The outcome is a biologically and chemically functional soil system that can cycle, release, and deliver nutrients when the tree actually needs them.



ORGANIC MATTER & CARBON
Fuel soil biology, improve structure, and increase nutrient holding capacity.

BIOLOGY & MICROBES
Drive nutrient cycling, break down organic matter, and make nutrients available to the tree.

PHYSICAL PROPERTIES
Good structure, porosity, and moisture balance allow roots and nutrients to move.

CHEMICAL BALANCE
Proper pH and nutrient balance ensure nutrients are in forms plants can access.

PRESENT IN THE SOIL ≠ AVAILABLE TO THE TREE
All the pieces work together to deliver nutrients to the root zone when the tree needs them.

FROM SOIL TEST TO PRESCRIPTION BLEND

How Hybrid-Ag dry blends turn soil data into root-zone delivery.

A prescription blend is not just fertiliser. It is a delivery system built around the soil, the crop, and the season.

01 START WITH THE SOIL TEST

A soil test shows what is present, what is missing, and what is out of balance. It gives the baseline for the blend.

We also consider pH, crop demand, block history, timing, and what the grower can realistically execute.



SOIL ANALYSIS SUMMARY

	Result	Rating
Phosphorus (P)	18 ppm	Medium
Potassium (K)	152 ppm	High
Calcium (Ca)	2100 ppm	High
Magnesium (Mg)	240 ppm	Medium
Sulfur (S)	12 ppm	Medium



02 BUILD THE PRESCRIPTION

Each blend is tailored to the crop, the block, and the production system. We match need, right form, rate, and timing.

Calcium, magnesium, phosphorus, potassium, sulphur, nitrogen, and trace elements are selected to suit the requirement.



03 WHY HUMAX 50 MATTERS

Humax 50 provides the carbon-rich biological base in many prescription blends.



High active carbon



Microbial diversity

HUMAX 50
ACTIVE CARBON COMPLEX



Holds soluble nutrition



Supports nutrient cycling



Metagenomic testing described the active carbon result as exceptionally high.

It helps create a blend that can hold, buffer, feed, and release nutrients more effectively through the root zone.

04 DELIVER THROUGH THE ROOT ZONE

The value is not only what goes into the blend. It is how the blend behaves once applied.



Hold nutrients in the treated zone



Support biological cycling



Improve release in time with demand



Help move nutrition from 'present on paper' to available in the root zone



KEY TAKEAWAY

Prescription dry blends bridge the gap between the soil test and the tree response. They are built to hold, cycle, release, and deliver nutrients when the tree needs them.

Misconception #5 - “One Leaf Test Tells the Story”

YOUNG LEAF

Less-mobile nutrient deficiency shows first:

Calcium, boron, zinc, trace elements

NUTRIENT TRANSPORT (REMOBILISATION)

Mobile nutrients are remobilised from older leaves into new growth.

OLD LEAF

Mobile nutrient deficiency shows first:

Nitrogen, phosphorus, potassium, magnesium, molybdenum



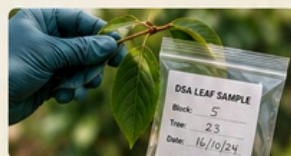
Nutrient issues do not all show in the same part of the tree.
So one leaf position can only tell part of the story.



MOBILE NUTRIENTS like nitrogen, phosphorus, potassium, magnesium and molybdenum can move from older leaves into new growth. When supply is short, older leaves often show the first signs.



LESS-MOBILE NUTRIENTS like calcium, boron, and key trace elements behave differently. Once deposited, they are harder for the tree to move, so issues often show closer to newer tissue, developing fruit, and growing points.



That is why Differential Sap Analysis looks at both young and old leaves. It gives a clearer view of nutrient movement, availability, and imbalance while there is still time to respond.

By comparing both positions, DSA helps show not just what is present, but how nutrients are moving through the tree.

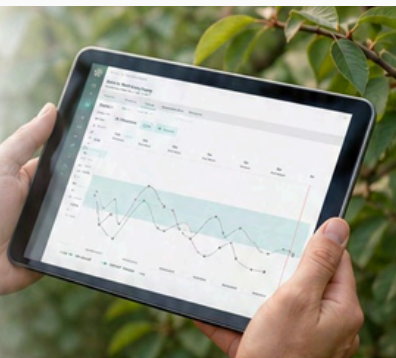


“Every nutrient in every tank we spray is there because the sap analysis has indicated it needs to be there.”

— **Mike Omeg**
Orchard View,
sweet cherry grower

What Is Your Data Telling You?

Frequent sampling, layered through **our nutrient software**, brings nutrient movement and plant response into focus — every key growth stage, every application, and every seasonal shift sitting on the same timeline.



ONE VIEW. EVERYONE ALIGNED.

Instead of isolated test results, the data tells a complete seasonal story. Everyone involved — grower, agronomist, and manager — sees the same view of where the program is working, where limitations are developing, and where adjustments are needed.



FROM DATA POINTS TO CLEAR INSIGHTS.

Patterns become clearer over time:

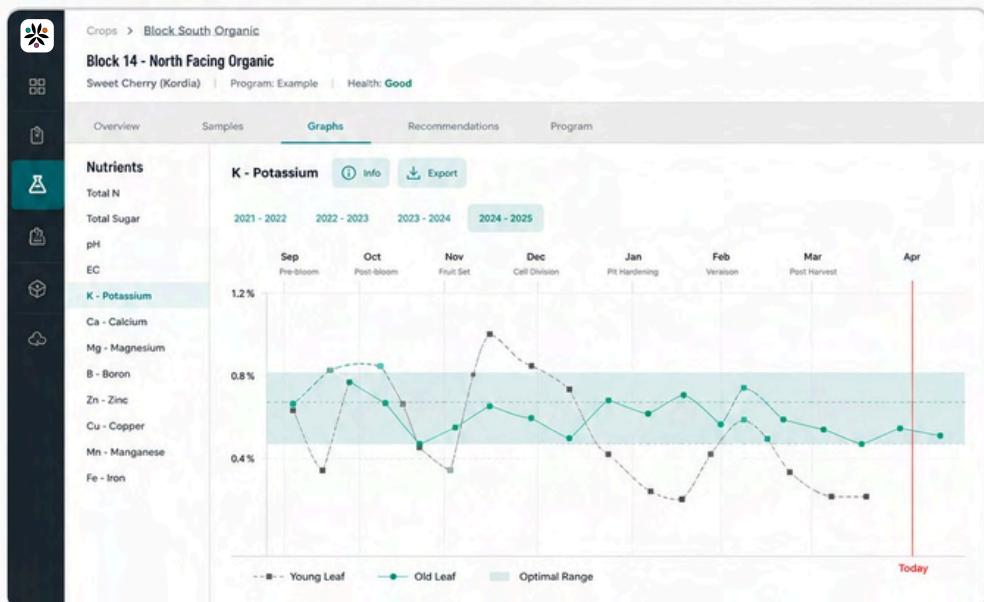
- how nutrients are moving
- where the plant is responding
- where deficiencies or imbalances are emerging
- how the crop reacts to management decisions through the season

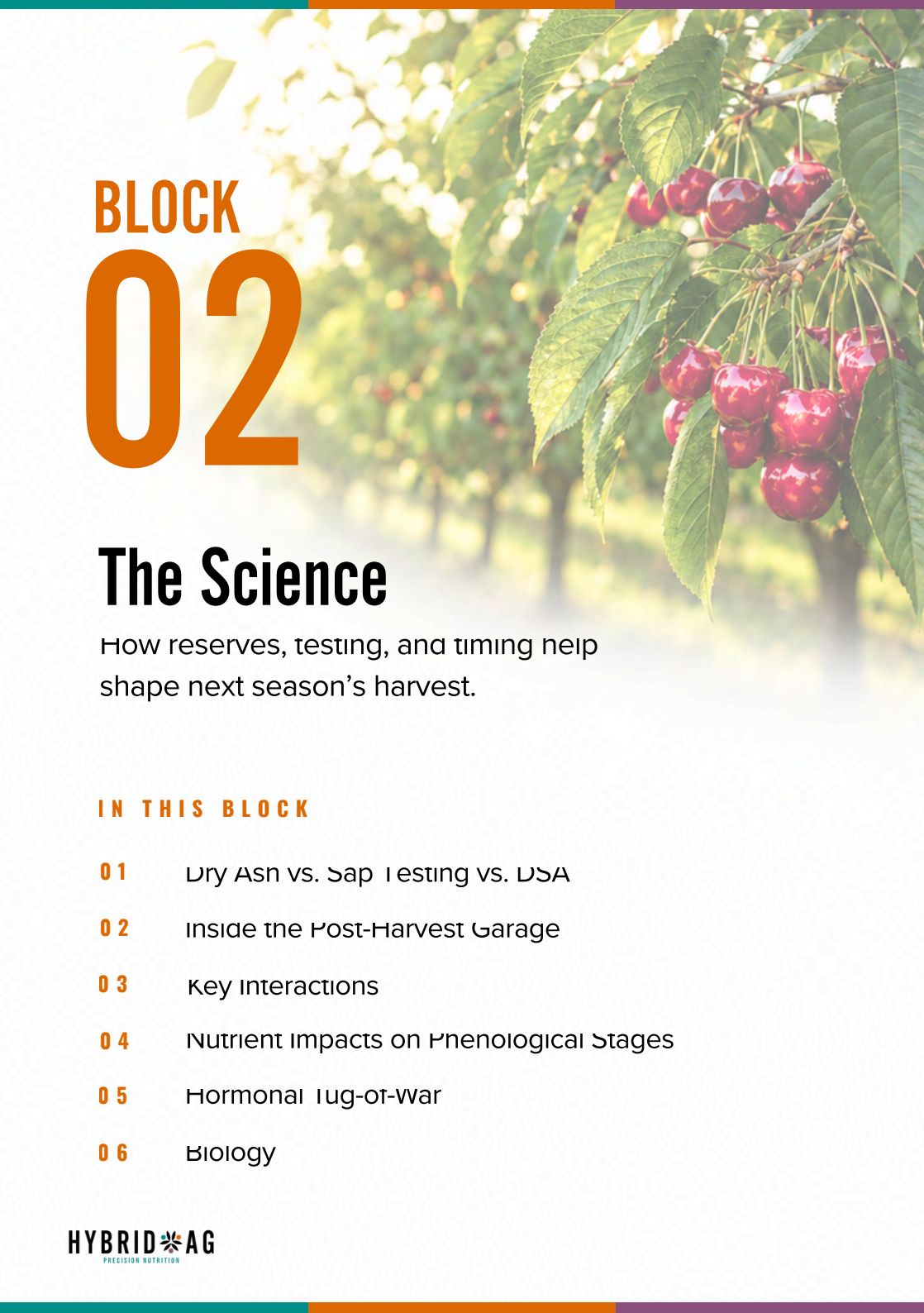


BUILDING BETTER PROGRAMS, SEASON AFTER SEASON.

Each season's lessons then carry into the next.

As more data is layered over time, the gap between the current program and the crop's ideal nutritional performance continues to close — block by block, season by season.





BLOCK 02

The Science

How reserves, testing, and timing help shape next season's harvest.

IN THIS BLOCK

- 01 Dry Asn vs. Sap Testing vs. DSA
- 02 Inside the Post-Harvest Garage
- 03 Key Interactions
- 04 Nutrient Impacts on Phenological Stages
- 05 Hormonal tug-of-war
- 06 Biology

PLANT TESTING

DRY ASH vs. SAP TESTING



DRY ASH

Tests the entire leaf; this includes cell walls which have a build-up of nutrients.

It can give false high's.



STANDARD SAP TEST

Extracts plant sap / cellular fluid from a sampled leaf to show what is available for immediate use in that sample.

A useful snapshot of available nutrition.



DIFFERENTIAL SAP ANALYSIS (DSA)

Extracts plant sap from young and old leaves separately to show what is available and how nutrients are moving through the plant.

Shows availability + mobility.

KEY DIFFERENCE	DRY ASH	STANDARD SAP TEST	DSA DIFFERENTIAL SAP ANALYSIS
Core reading	Whole-leaf content	Cellular fluid / plant sap	Cellular fluid / plant sap
Leaf samples	Usually one leaf type	Usually one leaf type	Young + old leaves
What it shows	What's in the tissue	What's available in that sampled leaf	What's available + moving
Nutrient movement	Limited visibility	Availability snapshot	Shows mobility / differential
Deficiency timing	Often after symptoms	Earlier warning	Earlier warning + differential insight

K - Potassium ppm

Dry Ash 4703
Standard SAP 5750
DSA Young Leaf 6862
DSA Old Leaf 3900



DIFFERENTIAL SAP ANALYSIS (DSA)

- Samples of old leaves and samples of young leaves.
- Sent off as separate samples but reported on one report as separate lines.
- Get more than double the information
 - ✓ Young leaf
 - ✓ Old leaf
 - ✓ The difference between young and old leaves is more powerful than raw numbers.



Learn More or Book a DSA Test

Scan the QR code to learn more.



THE SCIENCE

The Seasonal Race

"If you want to change next season, start the day harvest finishes."

— Andrew Smith, Hybrid-Ag

10 MATURATION

Fruit ripens and accumulates sugars and flavour.

9 CELL EXPANSION

Cells expand, driving fruit size increase.

8 CELL DIVISION

Rapid cell division increases fruit potential.

7 FRUIT SET

Pollinated flowers develop into small green fruit.

6 BLOOM

White cherry blossoms open, enabling pollination.

The Start/Finish

This stage marks the beginning of the season and the immediate start of the harvest.

Post-harvest isn't the end. It's the start of the next season.

Post-harvest isn't the end. It's the start of the next season.

POST-HARVEST
This stage marks the end of one season and the beginning of the next.

FINISH

Every stage counts. Start strong. Finish stronger.



/ START

1

HARVEST

marks the end of
and the immediate
g of the next.

t the end of the lap.
of the next one.

Line: Post-Harvest

the end of one season
beginning of the next.

t the end of the lap.
of the next one.



2

SENESCENCE (RECOVERY)

Critical stage where
the plant recovers and
prepares for dormancy.

3

DORMANCY

Growth stops.
The tree prepares
internally for winter.

4

QUIESCENCE

Buds are fully formed
but remain inactive
until conditions
are right.

5

BUD BREAK

Buds swell and
burst as growth
resumes.

THE SCIENCE

Inside the Post-Harvest Garage: Fo

Every part of the tree has a role to play after harvest.
Support the reserves. Protect next season's potential.



1. SPURS / THE FUEL TANK

Closest to the buds, where early spring reserves are first drawn on.

If this tank is low, bud strength, flower quality and early set can suffer.

WHY IT MATTERS

Spurs store carbohydrates that power bud break, flower development and early cell division.

CARBOHYDRATE RESERVES

Starch stored in roots + woody tissues.

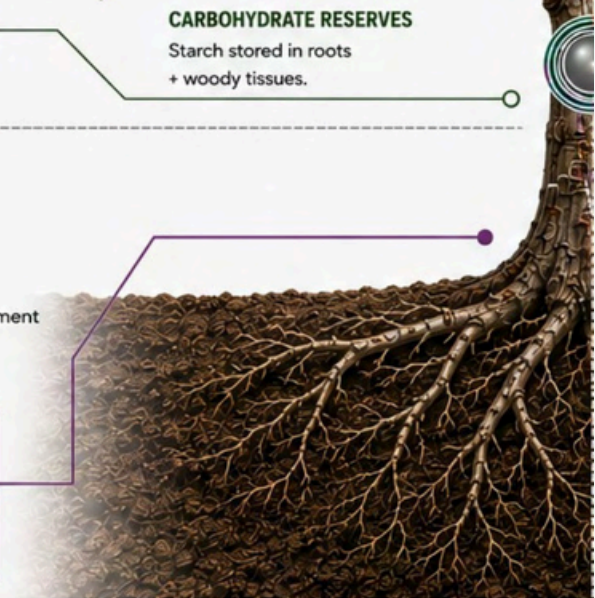


3. LOWER TRUNK / THE CHASSIS

The structural reserve bank. It connects the canopy and root system, supporting movement between above-ground and below-ground storage.

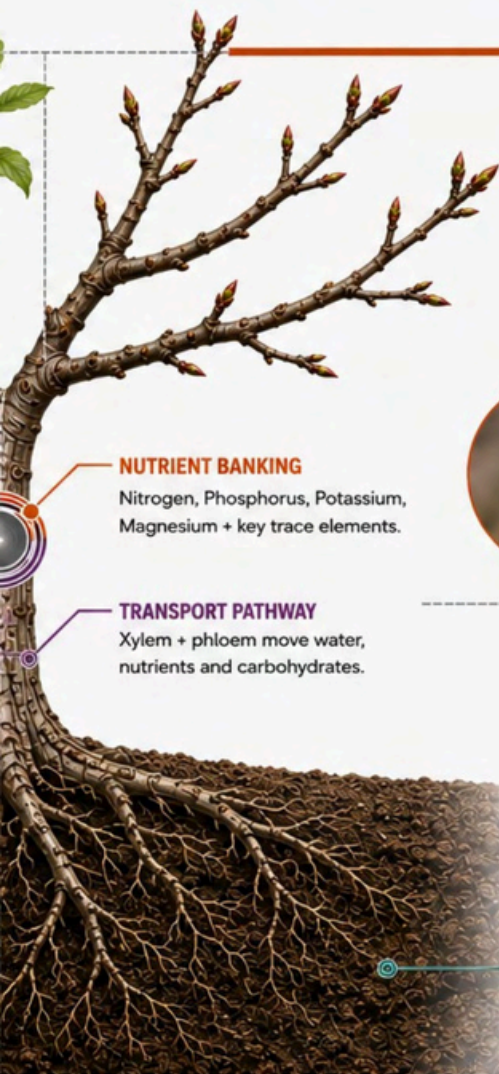
WHY IT MATTERS

The trunk stores and transfers reserves and provides structural strength. Damage here reduces the tree's ability to recover and perform.



When picking finishes, the tree rolls into the Post-Harvest Garage. Miss one bay, and the engine runs rough next spring.

ur Service Bays



NUTRIENT BANKING

Nitrogen, Phosphorus, Potassium, Magnesium + key trace elements.

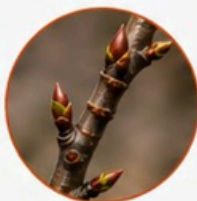
TRANSPORT PATHWAY

Xylem + phloem move water, nutrients and carbohydrates.

2. SCAFFOLD WOOD / THE ENGINE BLOCK

A mid-tree storage and distribution zone.

It helps bank and move reserves that support spring growth, flower development and early cell division.



WHY IT MATTERS

Scaffold wood determines how efficiently the tree can store, mobilize and allocate resources.

4. ROOTS / THE TYRES

Deep, long-term storage and the tree's connection to the soil.

A key zone for starch reserves and next-season recovery.



WHY IT MATTERS

Roots anchor the tree, absorb nutrients and water, and store the reserves needed to fuel next year's performance.

THE SCIENCE

Key Interactions

Antagonisms and synergies between cherry nutrients — read by stage, not in isolation.

There's a lot going on — that's why we test, not guess.

Post-Harvest

Replenish reserves, rebuild leaf area, and support bud development for next season.



Maturation

Fruit ripens. Quality, size and flavor develop.



Cell Division

Rapid cell division builds the potential for yield.



Fruit Set

Pollinated flowers develop into small green fruit.



ANTAGONISM

Reduced availability of one nutrient due to the influence of another.

STIMULATION

High level of a nutrient increases the demand by the plant for another nutrient.

Manganese (Mn)

Copper (Cu)

Phosphorus (P)

Molybdenum (Mo)

Bloom

Flowers open. Pollination and fertilization.



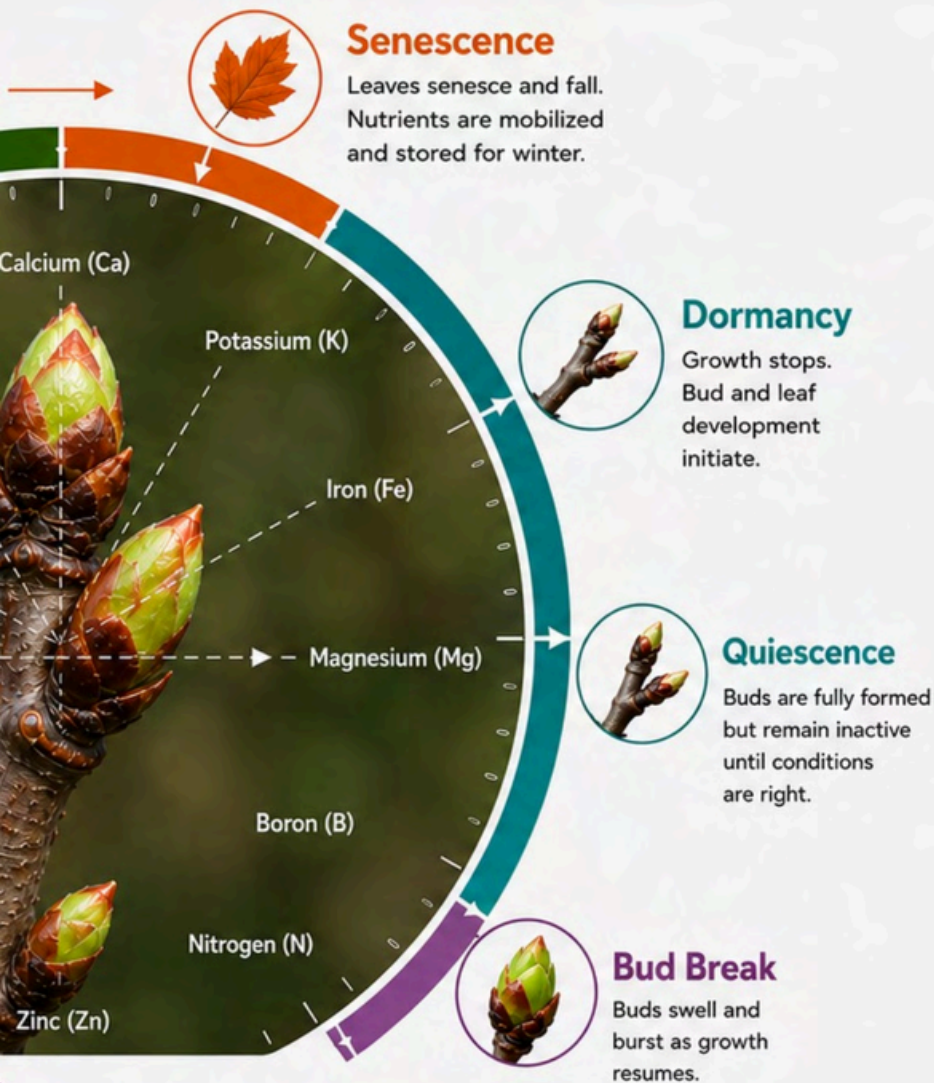
RATE

Enough to move the needle, not enough to create imbalance.



TIME

The biological window decides the result.



pen.
n and
n occur.



FORM

Chelated, soluble, foliar, or granular affects availability.



PLACE

Leaf, root zone, or soil profile.

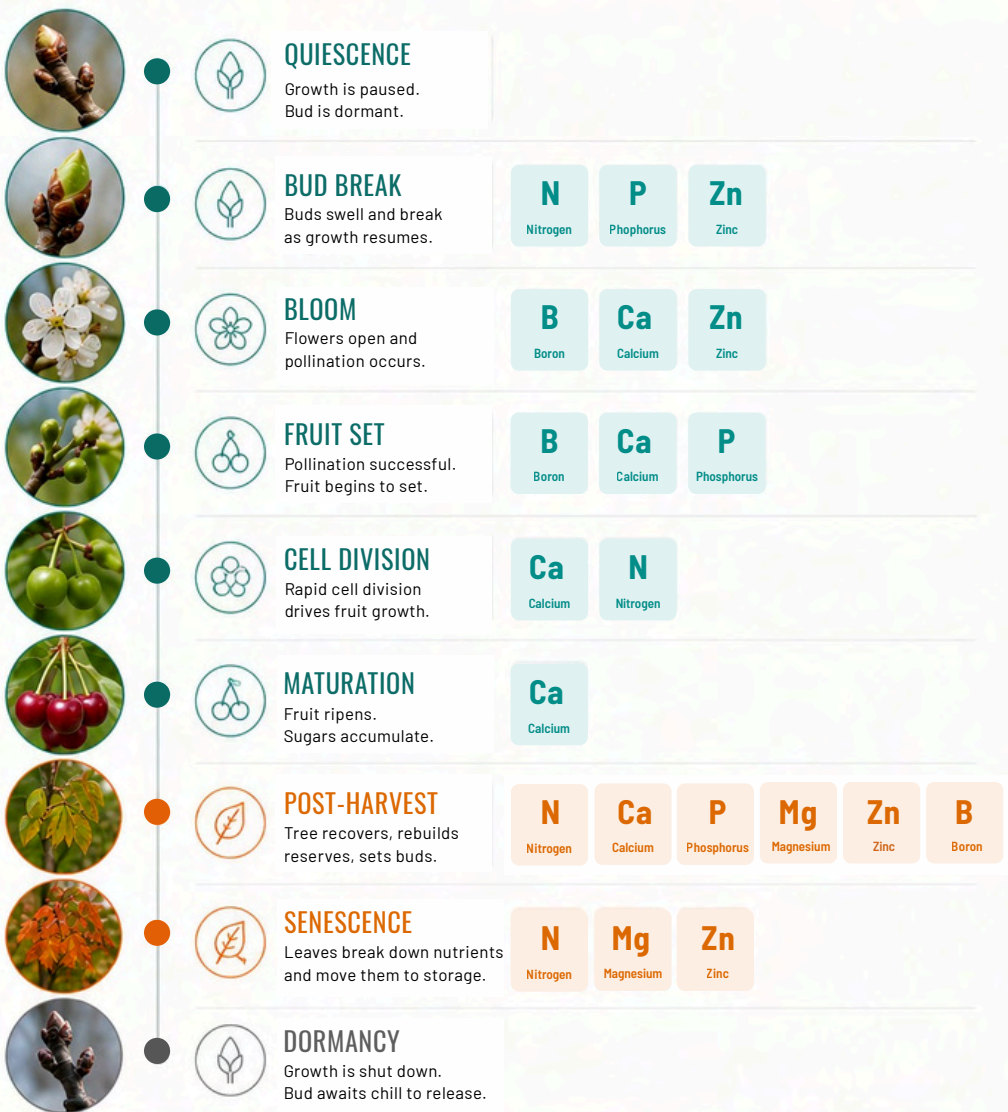


SOURCE

Choose the nutrient source and carrier that best fit the crop stage.

Nutrient Impacts on Phenological Stages

Which nutrients matter at each stage — and where your post-harvest decisions echo into next season



The **orange zones** are where post-harvest decisions echo into next season.



Strong nutrition after harvest builds stronger buds, better bloom, and higher yield.



Nutrient Roles

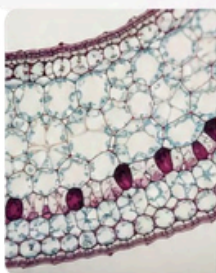
Every nutrient plays a specific role in the tree — especially during post-harvest and senescence.



N Nitrogen

Reserve loading,
amino acids

Post-Harvest + Senescence



Ca Calcium

Cell wall integrity,
fruit firmness

Post-Harvest → Cell Division



Zn Zinc

Bud development,
pollen viability

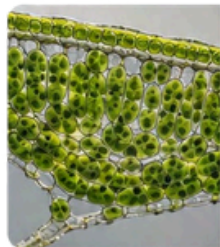
Post-Harvest + Senescence



P Phosphorus

Energy transfer
(ATP)

Post-Harvest



Mg Magnesium

Photosynthesis
support

Post-Harvest + Senescence



B Boron

Pollen tube growth
& fertility

Post-Harvest + Bloom



Right nutrient,
right time.



Post-harvest & senescence
decisions set next season's
potential.



Strong reserves =
stronger buds, better bloom,
higher yield.



THE SCIENCE

Hormonal Tug-of-War

A delicate chemical balance regulating leaf senescence and nutrient remobilisation.



BRAKES ON

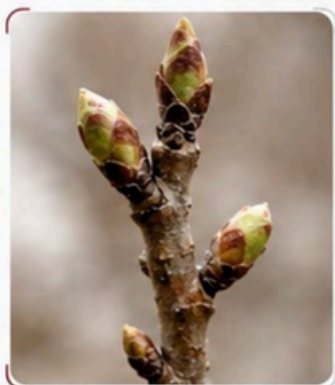


Promotes
Senescence



Ethylene & ABA

Ethylene signals leaf drop and chlorophyll breakdown. ABA supports stress response, dormancy preparation and programmed senescence. Together, they push the tree toward shutdown.



AUXIN

Moves down
from the shoot
to the roots.

Calcium

Calcium

The Calcium P

Calcium is a slow m
Early, active shoot g
is the key window
moving calcium i
new tissue.

As that window nar
timing and nutri
competition mat

To get the best t
out of calcium, it n
to be early. Once s
growth slows and
tree shifts toward
growth, calcium ca
be present, but tim
and competition m

TRIGGERS (Factors)



Age



Light



Ethylene



ABA



Temperature

LEVERS (What growers can influence)



Cytokinin



Auxin



Gibberellins



Nutrition



Stress Mgmt

BRAKES OFF



Delays
Senescence



Cytokinin, Auxin & Gibberellins

Maintain active growth signals, chloroplast function and canopy activity. Together, they help delay premature senescence and extend the photosynthetic window.

Pathway

over.
growth
w for
into

rows,
ent
ter.

use
needs
shoot
the
fruit
n still
ming
atter.



Timing is everything.

Apply calcium early while shoots are actively growing to capture the window.



Support the pathway.

Balanced nutrition and strong root activity improve calcium movement and uptake.



Balance the hormones.

Manage triggers and leverage growth hormones to delay senescence and protect canopy function.



Protect your potential.

Keep leaves healthy longer for more photosynthesis, better reserves and higher yield.

Biology



We farm above ground.
But the work happens underground.



Biology sits in the program for a reason. In the soil primer window, products such as **Mycro-Feast**, **Mycro-Tec SP**, and **Nutri-Core Soil Activator** are used just before spring to help wake the root system and support the tree as it moves into bud break and flower.



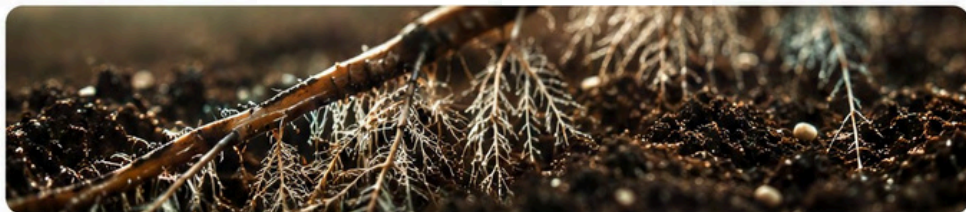
Biology can also support the broader program in-season. Around rain-risk windows, **biological sprays** may be incorporated before and after rain events to add another layer of support during plant stress events.



Biology works best as part of the **full program** — alongside good nutrition, an advised crop protection strategy and appropriate timing.



Healthy soil biology improves root function, nutrient availability, and stress tolerance — setting the foundation for **strong trees, quality fruit, and long-term productivity.**



HOW BIOLOGY SUPPORTS THE PROGRAM

THREE ROLES.

WORKING IN THE RIGHT WINDOW.

Biology plays a supporting role at key times to help your trees perform.



JUST BEFORE SPRING

01 SOIL PRIMER

Used to help wake the root system and support early root-zone activity as the tree moves into bud break and flower.

EXAMPLES:

Mycro-Feast • Mycro-Tec SP
Nutri-Core Soil Activator



BUD BREAK + FLOWER

02 ROOT-ZONE ACTIVITY

Supports root activity, nutrient cycling, and the early-season transition from dormancy into spring demand.



BEFORE AND AFTER RAIN

03 RAIN-EVENT SUPPORT

Biological sprays may be incorporated before and after rain events as part of the broader crop protection program, adding biological support during plant stress events.

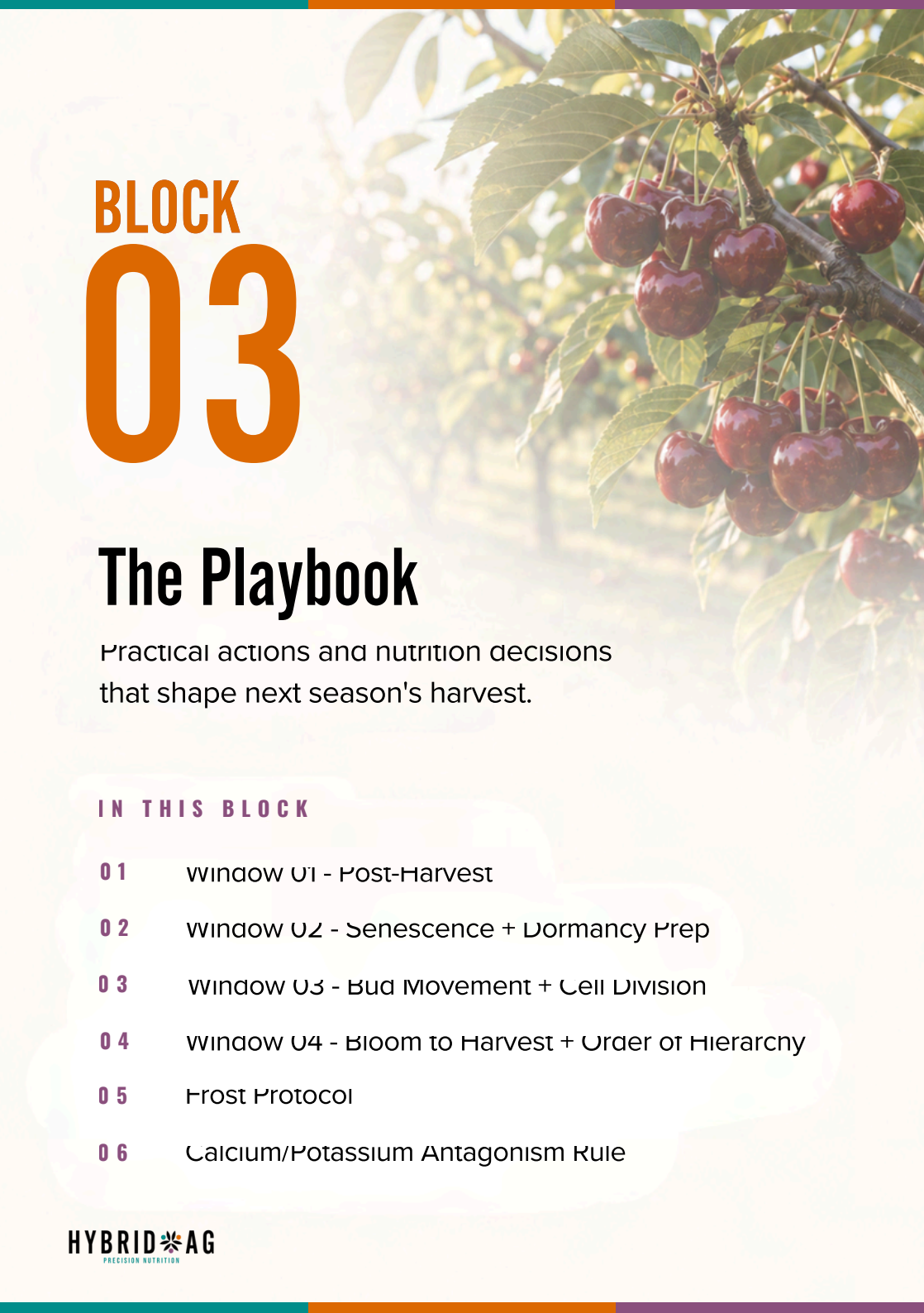
EXAMPLES:

Mycro-Tec SP • Micro 5000



Use in conjunction with the advised crop protection program.





BLOCK 03

The Playbook

Practical actions and nutrition decisions that shape next season's harvest.

IN THIS BLOCK

- 01 Window U1 - Post-Harvest
- 02 Window U2 - Senescence + Dormancy Prep
- 03 Window U3 - Bud Movement + Cell Division
- 04 Window U4 - Bloom to Harvest + Order of Hierarchy
- 05 Frost Protocol
- 06 Calcium/Potassium Antagonism Rule

The Post-Harvest Pit Stop

Fuel the Tree. Fill the Reserves. Set Up Next Season.



POST-HARVEST

Jan – Mar



MISS IT

- No program.
- Tree depletes reserves.
- Nutrients lost to orchard floor.



MASTER IT

- DSA-guided nutrition.
- Active reserves.
- Carbs maxed out.



SENESCENCE

Apr – May



MISS IT

- Premature leaf drop.
- Nitrogen and Phosphorus dumped, not exported.



MASTER IT

- Controlled shutdown.
- Efficient Nitrogen, Phosphorus, Potassium export to wood.



DORMANCY

Jun – Jul



MISS IT

- Empty tank.
- Depleted starch.
- Weak buds locked in.



MASTER IT

- Full tank.
- Maximum starch.
- Strong buds.
- Climate-ready.



SPRING

Aug – Nov



MISS IT

- Weak bud break.
- Poor set. Small fruit.
- Low pack-out.



MASTER IT

- Strong flowering.
- Full set.
- Premium size.
- Bankable consistency.



What you do after harvest determines what you harvest next.

Refuel the tree today with the right nutrients at the right time to build the yield, quality, and profitability of tomorrow.



Better reserves



Stronger performance



Higher returns

The Window to Refuel, Repair, and Prepare

PHASE 1 · BUILDING

POST-HARVEST

Green canopy, still photosynthesising.
Factory running at full capacity



Carb
production



Bud
initiation



Nutrient
loading

PHASE 2 · DISMANTLING

SENESCENCE

Autumn colours, exporting nutrients.
**Factory shutting down – shipping
 to warehouse**



Chloroplast
breakdown



Nitrogen &
Phosphorus
export



Leaf
drop



REFUEL

Load carbohydrate reserves into permanent wood and roots, your primary energy currency for spring



PREPARE

Strategic nitrogen timing to build amino acid reserves for spring, not pushing late veg growth.



REPAIR

Support cell wall integrity, strengthen buds and spurs. This is where calcium is critical.



PROTECT

Maintain canopy health to maximise the photosynthetic window while it's still open.

Post-Harvest Action Plan

PHASE 01: JAN – MAR POST-HARVEST

- Run a DSA test - young + old leaf sap to see reserves vs what's mobilised
- String DSA tests across the season to track nutrient drawdowns.
- Maintain irrigation - don't stress the canopy
- Fertigate & foliar apply nutrients to build reserves

PHASE 02: APR – MAY SENESCENCE

- Late-season urea for amino acid reserves and frost tolerance
- Foliar Magnesium to extend the photosynthetic window
- Foliar Zinc before leaf colour change for pollen viability
- Support programmed (not stochastic) senescence

PHASE 03: JUN – JUL DORMANCY PREP

- Reserves move in permanent wood and roots
- Starch accumulation peaks – battery charging
- 100% of spring energy from stored reserves
- Root uptake minimal during early dormancy phase

What happens if the fuel tank or the battery is empty?

Think of the tree like a battery. If it's empty, the season won't start. Spring runs 100% on stored reserves, not root uptake.

Post-harvest is your pit stop.



Autumn Soil Primer

Three moves while the leaves are still green.

01 SOIL APPLIED

Fungal inoculant + food source · soluble calcium

Supports residue breakdown and calcium bio-availability at bud break

02 FOLIAR REPLENISH

Boron+ zinc post-harvest sweep

Loaded into reserves before leaves drop

03 CARBON IN

Humic / fulvic in liquids

Nutrient Use Efficiency lever and feeds soil biology

*Get the start line right and the next growth stages compound.
Miss it and you fight next season uphill.*

Senescence + Dormancy Preparation



SENESCENCE SUPPORT

Senescence is active.

Yellowing leaves are not a failure. They are the tree pulling mobile nutrients out of leaves and shifting them into permanent storage.

This export is metabolic work, and you can support it.

WHAT TO DO	WHY
 Soil priming • carbon in	Support biology ahead of the quiet period
 Trace top-up • zinc + boron sweep	Support bud and trace status before leaf drop
 Reduce stress • antioxidants or biostimulants	Don't push — support the transition



DORMANCY PREP

A quiet tree is still moving internally.

Root uptake slows dramatically during dormancy, but the tree is not switched off. Sap mobility and internal redistribution can still continue as reserves, water, and nutrients move through woody tissue, buds, and storage organs. This window is about making sure the resources are already there so the tree can finish dormancy preparation and wake up to a primed pantry.

WHAT TO DO	WHY
 Soil blend macro corrections	Balance the pantry before spring demand
 Trace audit + corrections	Target gaps before bud movement
 Carbon + microbe food source	Prepare the root zone for release, cycling, and restart



KEY TAKEAWAY

Dormancy is quiet, not dead. Root uptake is low, but internal movement still matters. The job is to have the resources in place before the tree needs them.

The Mechanics of Senescence



Autumn leaves are not dying, they are being dismantled.

Nutrients are actively remobilised, degraded from the canopy, and permanently stored inside the plant's woody architecture.



**Leaf drop does not mean the tree is done,
it means the tree is packing its bags.**



POST-HARVEST PHASE

Resources are redirected from fruit to the tree. The focus is on rebuilding and storing.



SENESCENCE PHASE

Leaves dismantle and export nutrients. The focus is on transfer and storage.



Best results come from understanding these phases.

Bud Movement + Cell Division



BUD INITIATION

Jan - Feb

Meristem active
Buds visible
and growing



FLORAL DIFFERENTIATION

Feb - May

Sepals, petals,
stamens,
pistil forming



DORMANT

Jun - Jul

Scales closed
Reserves locked
Waiting



SPRING BLOOM

Aug - Sep

Everything built
post-harvest, quality
is locked in



+22%

BUD NUMBERS

More reserves =
more viable primordia initiated



BUD SIZE

Larger buds =
stronger flowers & pollen



FLORAL BALANCE

Adequate carbs favour
fruiting over veg



*Next Season's Bloom Quality is set during Post-Harvest —
not next August.*

Cell division decides cell number



After fruit set.

Every cell that will ever exist in this season's fruit is created in this window.

Cell number shapes much of what follows.

There is a 10–14 day window after fruit set when much of the fruit's final cell number is being built. Calcium availability in this period strongly influences cell-wall integrity, firmness, and shelf life. Miss this cell division window and we can't correct this at a later date.



01 BUD MOVEMENT

Use nitrogen in the right form and timing as bud movement begins. Soil priming supports the shift into early reproductive growth.



02 BUD + FLOWER

Foliar calcium + boron + zinc around flower supports early reproductive demand and new cell-wall development.



03 CELL DIVISION

10–14 day window after fruit set. Front-load calcium early. Potassium becomes more important later, as cell expansion and fruit fill begin.



Right inputs. Right timing. Right window.

That's how you build better fruit.

Cell Expansion / Fruit Fill



From bloom to harvest.

The Potassium (K) window opens.

Royal Tioga 60–65 • Samba/Lapin 65–85

Once cell number is set, the focus shifts to fruit expansion, sugar loading, colour, and finish. This is where potassium becomes the dominant lever. Manganese, iron, and copper help support photosynthesis and leaf function so the tree can keep filling fruit. At this stage, quality becomes the commercial test — size, firmness, colour, and pack-out.

In a short cherry season, consistency matters.

FRUIT FILL STACK

K	Potassium (K)	Sugar transport, colour, size
Mn	Trace Suite	Manganese • Iron • Copper
	Carbon in	Humic / fulvic support for nutrient efficiency through fill



ORDER OF HIERARCHY

How You Apply Matters as Much as When

Soil, foliar, fertigation — same nutrient, different result.

By this stage, calcium still matters — but how it is applied matters more. Ground balance and fertigation do the heavy lifting, with foliar used as support where needed. Potassium becomes the main fruit-fill lever, while trace nutrition supports photosynthesis and canopy function.



BEST BANG BY NUTRIENT

Ca

Calcium (Ca)

Soil • ground balance first,
then foliar fill-in

K

Potassium (K)

Soil + fertigation post-cell division.
Foliar in late fill.

B+Zn

Boron + Zinc (B + Zn)

Foliar at flower.
Soil for trace audit.

N

Nitrogen (N)

Soil ammonium at bud movement.
Not at frost.



Right input, right timing, right window

Frost is an osmotic injury — *not just a freeze.*

THE MECHANISM



01 EXTRACELLULAR ICE

Water outside the cell
(lower solutes) freezes first.



02 OSMOTIC PULL

Gradient drags water out of cells
across the plasma membrane.



03 DAMAGE CASCADE

Wilting · lysis on thaw · xylem
embolism · ROS · protein
denaturation.

THE NUTRIENT LEVERS



PHOSPHORUS (P) ENERGY

ATP fuels cellular repair
and defence activation
under stress.



CALCIUM (CA) MESSENGER

Cell-wall integrity and
secondary messenger that
switches on cold-response
genes.



SILICON (SI) WALL ARMOUR

Stronger walls, regulates
water flow during
osmotic shock.



POTASSIUM (K) BUFFER

Osmotic balance, citrate-form
antioxidants, stomatal
regulation, cryoprotectant
synthesis.



FROST-EX PROTOCOL



**24–48 HOURS
BEFORE**

Applied pre-event



LIQUID K 35%

K citrate → antioxidants



SUGAR SYRUP

Antifreeze proteins ·
freeze-pt depression



FULVIC

Cryoprotectant ·
water binding



BLACK KELP (KELP)

Trace + amino acids ·
sugar accumulation



Why not Potassium Nitrate at Frost

Nitrate pushes vegetative growth — new tissue is NOT cold tolerant. Disrupts solute balance. Metabolic shift away from defence (plant burns energy converting nitrate + amino acids). Faster water uptake = faster ice formation.

“

During a frost event we want the plant focused on its natural defence mechanisms. An influx of nitrate ions causes a metabolic shift — the plant tries to convert nitrates to amino acids and takes its focus off protecting itself.

Right steps, taken at the right
time, **create good outcomes**
and options in the market



Small decisions, timed right,
change the whole outcome.

What's up for grabs



CONSISTENT YIELDS

Crop loads that hold together through variable weather, lower chill, and warmer seasons.



FRUIT QUALITY: FIRMNESS, SIZE, AND BRIX

A quality stack that helps fruit make grade and hold up through packing, shipping, and sale.



REPEAT SALES

Cherry season is short. A good eating experience brings customers back; one poor batch can lose them until next year.



MORE MARKET OPTIONS

Better size and firmness grades open more market channels such as export, domestic, supermarket — segregation power.



SCAN FOR A FREE NUTRIENT AUDIT

You're sitting on more than you think.

Most growers have years of soil tests, leaf results, water analyses and yield records spread across files and inboxes.

We're offering a free Nutrient Audit — we pull it all together, overlay it, and show you the moves that are actually up for grabs.



Scan me!

SAMPLE NUTRIENT MANAGEMENT PROGRAM

CHERRY NUTRITION PROGRAM

PHENOLOGICAL STAGES		DORMANCY	BUD & FLOWER AND CELL DIVISION				
CROP STAGES		Dry Ground Spread	Early Bud Break	20–30% Flowering	80–90% Flower	Shuck Fall	Fr Appl
							
APPLICATION TIMING		August	August	September	September	September	Oct
TESTING				DSA	DSA	DSA	D
 SOIL APPLICATIONS	Mycro-Feast®	10 L					
	Nutri-Core Soil Activator	75 L					
	Mycro-Tec SP	10 L					
 FOLIAR APPLICATIONS	Power-Cal®		10 L	10 L	10 L	10 L	1
	Opti-Trace® Bud Max		5 L	5 L	5 L	5 L	
	Bio-See® F-KF		2.5 L	2.5 L	2.5 L	2.5 L	2
	Agri-Vive® RF					10 L	10
	Opti-Trace® Bio-Check					2 L	2
	Opti-Trace® Max						
	Agri-Vive® 0.0.30						
	Opti-Trace® Boron						
	Opti-Trace® Copper						
	Power-N®						
 FERTIGATION	Opti-Cal®		30 L	30 L			20
	Agri-Vive® RF					20 L	20
	K-Max						
	Nutri-Core® Multi N						

FRUIT DEVELOPMENT & CELL EXPANSION						POST-HARVEST	
Fruitlet Application 1	Fruitlet Application 2	Fruitlet Application 3	Fruit Fill Application 1	Fruit Fill Application 2	Fruit Fill Application 3 (Later Varieties)	Post Harvest 1	Post Harvest 2
							
October	October	October	November	November	December	February	March
DSA	DSA	DSA	DSA	DSA	DSA		
0 L	10 L	10 L	10 L	10 L	10 L	10 L	10 L
			5 L	5 L	5 L	5 L	5 L
5 L	2.5 L	2.5 L	2.5 L	2.5 L	2.5 L	2.5 L	2.5 L
0 L							
0 L							
	3 L	3 L					
	5 L	5 L	5 L	5 L	5 L		
			3 L	3 L	3 L		
						0.3 L	0.3 L
						10 L	10 L
0 L	20 L						
0 L							
			20 L	20 L	20 L		
						40 L	40 L

How the Program Comes Together

A cherry nutrition program works best when it is built around the tree's changing demand through each phenological stage.

Post-harvest focuses on rebuilding carbohydrate and nutrient reserves for the following spring. Bud movement, flowering, and cell division are about supporting early reproductive demand and establishing the fruit's potential. Fruit development then shifts toward maintaining nutrition that supports size, colour, firmness, sugars, and pack-out consistency.

Each stage asks different things from the tree — and the program needs to move with it.



POST-HARVEST

Rebuild reserves for the next season.



BUD MOVEMENT

Support early growth and reproductive demand.



FLOWERING & CELL DIVISION

Establish fruit potential and set the foundation.



FRUIT DEVELOPMENT

Maintain nutrition for size, colour, firmness, sugars, and pack-out consistency.



The goal is not simply to apply more nutrition, but to match the right nutrient, in the right form, at the right timing, to the outcome you are trying to improve.



Key Players in the Program

Targeted nutrition for every critical window
in your cherry season.

Scan for more information or talk to us
about how it fits your block.



POST-HARVEST



Power-Cal

FOLIAR CALCIUM

Foliar calcium loaded while
the canopy is still active —
banked into next season's
buds and reserves.



Builds bud strength



Improves cell integrity



Supports better
flower quality



LEARN MORE



Opti-Trace BudMax

TRACE ELEMENT CHELATE

Chelated trace elements stored
into next season's bud —
driving uniform bud burst
and flowering.



Supports pollen
viability



Enhances bud
development



Optimises
flowering



LEARN MORE



Bio-Sea FKF

SEAWEED & FULVIC SUPPORT

Fish, kelp and fulvic carriage
through the senescence pass
while nutrients move to wood
and root storage.



Enhances nutrient
uptake



Improves calcium
transport



Supports flower
quality



LEARN MORE





DORMANCY / SOIL PRIMER



Mycro-Feast

SOIL BIOLOGY PRIMER

A balanced fungal and bacterial food source — wakes microbial activity and prepares soil for spring growth.



Stimulates microbial activity



Enhances nutrient cycling



Prepares soil for spring growth



LEARN MORE



Nutri-Core Soil Activator

SOIL ACTIVATOR

Supports soil-applied nitrogen and sulfur delivery, helps shift the rhizosphere, and supports calcium and trace element availability around the root zone.



Improves nutrient availability



Solubilises calcium & trace elements



Supports root zone health



LEARN MORE



Mycro-Tec SP

BIOLOGICAL INOCULUM

A biological inoculum used to support soil biology, root activity, and nutrient availability in the root zone.



High-count beneficial microbes



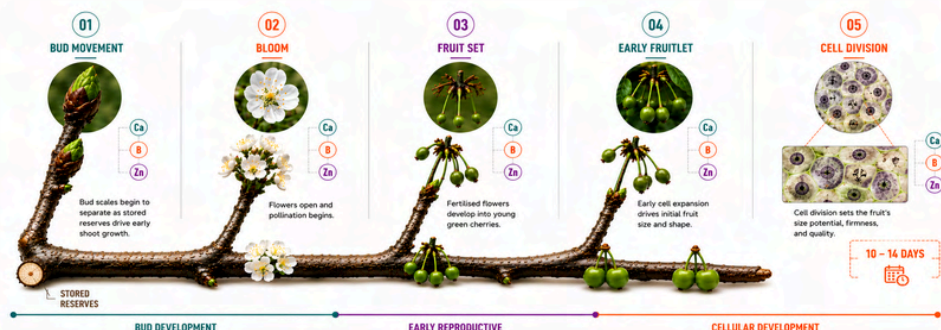
Supports root activity



Unlocks soil nutrient reserves



LEARN MORE



BUD & FLOWER AND CELL DIVISION



Power-Cal

FOLIAR CALCIUM

Calcium support through the bud, flower, and early cell-division window.



Supports cell division



Improves flower set & retention



Builds firmness & shelf life



LEARN MORE



Opti-Trace BudMax

TRACE ELEMENT CHELATE

Boron, zinc, and trace support around bud and flowering stages.



Supports pollen viability



Enhances bud development



Optimises flowering



LEARN MORE



Bio-Sea FKF

SEAWEED & FULVIC SUPPORT

Fish, kelp, fulvic, and plant-stimulant support alongside foliar passes.



Enhances nutrient uptake



Improves calcium transport



Supports flower quality



LEARN MORE



Agri-Vive RF

ACTIVE CARBON & NUTRIENT SUPPORT

Liquid nutrient and carbon support around shuck fall and early fruitlet demand.



Supports fruitlet demand



Improves nutrient utilisation



Strengthens early fruit development



LEARN MORE



FRUIT DEVELOPMENT & CELL EXPANSION



Power-Cal

FOLIAR CALCIUM

Ongoing calcium support where the program calls for it, especially where fruit quality and cell-wall support remain part of the plan.



Supports cell wall strength



Reduces split and soft fruit



Improves shelf life



LEARN MORE



Bio-Sea FKF

SEAWEED & FULVIC SUPPORT

Fish, kelp, and fulvic support through repeated foliar passes.



Enhances nutrient uptake



Improves transport and efficiency



Supports consistent fruit fill



LEARN MORE



K-Max

POTASSIUM

Potassium support through fruit fill, size, colour, and sugar loading.



Increases fruit size



Enhances colour development



Improves sugar loading



LEARN MORE



Agri-Vive RF

CARBON & NUTRIENT SUPPORT

Liquid nutrient and carbon support through in-season crop demand.



Supports plant metabolism



Improves nutrient use efficiency



Backs in-season demand



LEARN MORE

PRESCRIPTION BLENDS

Custom dry blends built around the soil, the crop, and the season.



CUSTOM DRY-BLEND SYSTEM



01 BUILT FROM THE SOIL TEST

- Starts with the baseline
- Identifies what is present, missing, or out of balance
- Tailored to the block



02 MATCHED TO THE SYSTEM

- Right nutrients
- Right forms
- Right rate and timing
- Built to suit application reality



03 HUMAX 50 FOUNDATION

- Active carbon complex
- Supports microbial activity
- Helps hold soluble nutrition
- Supports nutrient cycling



04 ROOT-ZONE DELIVERY

- Hold
- Buffer
- Feed
- Release
- Deliver nutrients where roots can use them



ROOT-ZONE DELIVERY

Nutrients are held, cycled, released, and delivered where roots can use them.



Not just fertiliser. A delivery system that helps hold, cycle, release, and deliver nutrients when the tree needs them.

MEET THE TEAM



Andrew Smith

Senior Nutritional Agronomist

Andrew is a generational cherry grower at Smiths Fruit in Taminick, Victoria, and leads Hybrid-Ag's horticulture team. He brings a grower's perspective to nutrition, combining orchard experience, field observation, and data-driven programs to help growers make stronger seasonal decisions.



Chelsea Hall

Nutritional Agronomist

Chelsea works closely with growers to translate soil, leaf, and crop data into practical nutrition strategies. With a strong grounding in plant physiology and soil biology, she helps connect what's happening in the system to what decisions need to be made in the field.



Nathan Strawbridge

Nutritional Agronomist

Nathan supports growers across horticulture and viticulture, with a focus on aligning nutrient timing with crop demand through the season. He works at the block level, helping growers interpret testing, adjust programs, and make decisions that improve consistency, fruit quality, and overall system performance.



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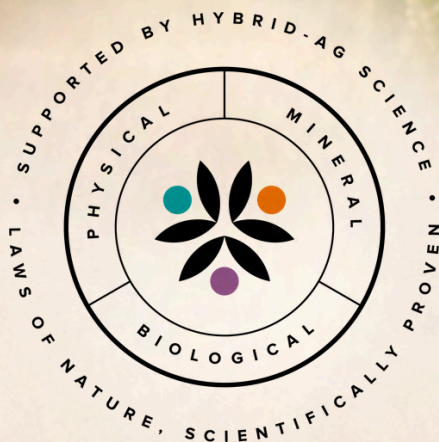
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**STRONGER DATA.
STRONGER DECISIONS. BETTER RESULTS.**

*Scan
me!*





OPTIMISING INPUTS  MAXIMISING OUTCOMES

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